Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

Mental Health Prediction Using Machine Learning

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Introduction

Mental health plays a vital role in maintaining emotional, psychological, and social well-being. In recent years, mental health disorders such as depression, anxiety, and stress have increased rapidly due to lifestyle changes, academic pressure, workplace stress, and social isolation. Traditional diagnosis methods are often time-consuming, expensive, and dependent on clinical experts. The advancement of Artificial Intelligence and Machine Learning has enabled the development of intelligent healthcare systems capable of analyzing large datasets and identifying mental health patterns efficiently. This research proposes a machine learning-based mental health prediction system that uses hybrid machine learning techniques to improve prediction accuracy and support early healthcare intervention.

Objectives

The main objective of this research is to develop an intelligent mental health prediction system using machine learning algorithms. The system aims to analyze behavioral and psychological data for identifying mental health risks at an early stage. Another objective is to compare the performance of different machine learning algorithms such as Logistic Regression, SVM, Random Forest, and Gradient Boosting. The research also focuses on improving prediction reliability using hybrid learning approaches and providing decision support for healthcare professionals. Additionally, the system aims to reduce manual effort and improve the efficiency of mental health assessment processes.

Related Work

Several researchers have applied machine learning techniques for mental health prediction and healthcare analytics. Earlier studies mainly focused on algorithms such as Support Vector Machine, Neural Networks, and Random Forest for detecting depression, anxiety, and stress-related disorders. Although these systems achieved moderate to high accuracy, many suffered from limitations such as high computational cost, low scalability, and insufficient feature optimization. Recent research highlights that ensemble and hybrid learning methods provide better stability and improved prediction reliability. The proposed system overcomes these challenges by combining multiple machine learning approaches to achieve higher accuracy and better generalization performance.

Dataset Description

The dataset used in this research contains behavioral, psychological, and demographic information collected from structured healthcare records. Important attributes include age, gender, sleep duration, stress level, anxiety level, academic pressure, social interaction, lifestyle habits, and medical history. The dataset consists of 5,000 records, where 80% of the data is used for training and 20% for testing purposes. Data preprocessing techniques such as missing value handling, duplicate removal, feature scaling, normalization, and label encoding were applied to improve the quality and consistency of the dataset before training the machine learning models.

Data Preprocessing

Data preprocessing is an important step in improving the performance and accuracy of machine learning models. In this research, preprocessing techniques were applied to clean and prepare the dataset before model training. Missing values and duplicate records were removed to maintain data consistency and quality. Label encoding was used to convert categorical values into numerical form for machine learning compatibility. Feature scaling and normalization techniques were applied to standardize the dataset and reduce variations among attributes. These preprocessing operations helped improve model efficiency, reduce errors, and enhance prediction reliability.

System Workflow

The proposed system follows a structured workflow for mental health prediction using machine learning techniques. Initially, healthcare-related datasets are collected and processed using data cleaning and preprocessing methods. After preprocessing, feature engineering and normalization techniques are applied to improve model efficiency. The processed dataset is then divided into training and testing sets for model development. Multiple supervised learning algorithms including Logistic Regression, SVM, Random Forest, and Gradient Boosting are trained and evaluated. Finally, the system generates prediction results along with performance visualization and analysis reports.

Results and Discussion

The experimental analysis demonstrates that the proposed hybrid machine learning model achieved superior performance compared to traditional algorithms. Logistic Regression achieved an accuracy of 84%, while SVM reached 87% accuracy. Random Forest and Gradient Boosting further improved the prediction accuracy to 91% and 92% respectively. The proposed hybrid model achieved the highest accuracy of 95% along with improved precision, recall, and F1-score values. The results indicate that ensemble learning methods effectively reduce overfitting and enhance prediction reliability. The analysis also revealed that factors such as high stress levels, reduced sleep duration, and low social interaction strongly contribute to increased mental health risk.

Limitations and Future

The proposed system depends on the quality and size of the healthcare dataset for accurate prediction. Mental health conditions vary among individuals, which may affect prediction reliability. The current model uses structured data and does not include real-time emotional analysis. Hybrid machine learning models may also require higher computational resources. In the future, deep learning and real-time monitoring systems can further improve prediction accuracy. The system can also be developed into a web or mobile healthcare application for wider accessibility .

Conclusion

This research presents an ML-Based Mental Health Prediction System using hybrid machine learning techniques for early detection of mental health conditions. The proposed approach integrates multiple supervised learning algorithms with preprocessing and feature optimization methods to improve prediction performance. Experimental results show that the hybrid model achieves high accuracy and reliability compared to conventional approaches. The system can assist healthcare professionals in early diagnosis and decision-making while reducing dependency on manual assessments. Future enhancements may include deep learning integration, real-time healthcare monitoring, and deployment as a web-based healthcare support system.

References

Smith et al. proposed a depression detection system using Support Vector Machine techniques for healthcare analysis. Johnson and Lee developed a neural network-based anxiety prediction model using behavioral data analysis. Kumar et al. implemented a Random Forest-based mental health classification framework to improve prediction performance. Additional references were collected from IEEE research papers, healthcare analytics journals, artificial intelligence studies, and machine learning documentation related to mental health prediction systems.

Thank You